

Customer Churn Prediction

Using Machine Learning on Telecom Subscriber Data

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Introduction

Customer churn is when a customer cancels their subscription or switches to a different company. This is a big problem for businesses that depend on subscriptions, like phone companies, streaming services, and software companies. It costs much more money to find a new customer than to keep one you already have. That's why it's so valuable to spot customers who might leave before they do.

Many things can cause a customer to leave, such as price, contract length, service quality, and personal background. These factors often mix in complicated ways, so simple rules can't always predict who will churn. Machine learning can study past customer data to find these hidden patterns. It can flag customers who are likely to leave, giving the company time to offer them a deal to stay.

Business Problem

Lowering churn is a top priority for any subscription business, since keeping a current customer is usually much cheaper than finding a new one. The people affected by churn prediction include:

- Subscribers and account holders
- Customer retention and support teams
- Marketing teams that create loyalty and win-back offers
- Product teams deciding which services or prices to change
- Company leaders tracking revenue and customer value

Importance of Churn Prediction

Churn directly hurts revenue. Losing a customer means losing not just their current payments, but also all the money they would have spent in the future. Older approaches to keeping customers often rely on broad promotions for everyone or only reach out after a customer has already canceled. Machine learning lets a company act early. It can score customers by churn risk in real time, so retention efforts focus on the customers who need them most.

Approach

Data Collection: This project will use the Telco Customer Churn dataset, which is free to download on [Kaggle](#). It has records for about 7,000 customers of a phone company, including their background, account details, services, billing information, and whether they churned.

Data Cleaning and Preprocessing: Fix missing values, correct data types, turn categories into numbers, and deal with the imbalance between "churned" and "stayed" customers using methods like SMOTE or class weighting.

Exploratory Data Analysis: Create charts to see churn rates by contract type, tenure, payment method, and service bundle, to find out which groups of customers are most likely to leave.

Feature Selection: Use correlation analysis and feature importance tools to find which factors affect churn the most and create new features when helpful (like average monthly spend or tenure groups).

Model Selection: Compare several types of models, including Logistic Regression, Random Forest, Gradient Boosting, and Support Vector Machines.

Training and Testing: Split the data into training and test sets, adjust model settings using cross-validation, and test the model on data it hasn't seen before.

Model Evaluation: Measure how well the model works using accuracy, precision, recall, F1 score, and AUC-ROC. Pay extra attention to recall, since missing an at-risk customer is costly.

Timeline

The project is expected to be finished in four (4) weeks, structured as follows:

Week	Milestone
Week 1	Collect the data, clean it, and explore it
Week 2	Engineer and select features, and handle class imbalance
Week 3	Build, train, and tune the model
Week 4	Evaluate the model, write up findings, and put together the final report/presentation

Progress Monitoring

Progress will be tracked by breaking the project into clear stages:

- Data collection and cleaning: tracked via missing values and data quality
- Exploratory analysis: tracked by finishing the planned charts (churn by contract type, tenure, charges, etc.)
- Modeling: tracked by comparing evaluation scores across different versions of the model
- Reporting: tracked against a shared outline, with each team member responsible for certain sections

Weekly check-ins will confirm that each stage is on track, and tasks will be reassigned if a team member is stuck or falling behind.

Deliverables

- 10-minute presentation video
- GitHub Repository with a polished README

Composition

Team Member	Role	Primary Responsibility
Jonathon Kahn	Team Lead	Coordinates the project, checks model, writes the final report
Atlas Li	Data Lead	Cleans and prepares the data, builds new features
Alexander Keifer	Modeling Lead	Picks, trains, and tunes the model
Deji Ajani	Tech Lead	Provides leadership in coding tasks and ensuring code quality